Summary Tables of `MAXCUT` Data

1 Tables for depth $P = 1$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Interp (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fourier	61 [10–20]	10 [12]	699 [10–20]	570 [10–20]
Fixed Angle	55 [10–20]	-	-	289 [10–20]
Fixed Angle*	55 [10–20]	-	-	273 [10–20]
Interp	83 [10–20]	10 [12]	805 [10–22]	382 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	65 [10–20]	-	592 [10–17]	280 [10–20]
TQA*	63 [10–20]	-	584 [10–17]	60 [10–20]
Recursive TS	81 [10–20]	10 [12]	481 [10–22]	368 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
		MPS		
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Interp (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		PP		
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3
TQA	-	-	-	-
TQA*	-	-	-	-
		SV		
Fourier	78.8 ± 3.1	79.3 ± 3.0	74.8 ± 4.2	82.0 ± 4.1
Fixed Angle	78.4 ± 2.6	-	-	80.7 ± 3.1
Fixed Angle*	79.1 ± 2.9	-	-	81.7 ± 4.3
Interp	78.8 ± 2.9	79.3 ± 3.0	74.6 ± 3.9	81.9 ± 4.2
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	77.6 ± 2.6	-	71.0 ± 5.1	79.0 ± 2.9
TQA*	78.8 ± 3.0	-	74.7 ± 3.6	78.1 ± 2.1
Recursive TS	78.8 ± 2.9	79.3 ± 3.0	75.0 ± 4.8	82.0 ± 4.3

Table 2: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	R
MPS								
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle (Aer)	-	-	-	-	-	-	-	
Fixed Angle (Aer)⋆	-	-	-	-	-	-	-	
Fixed Angle	-	-	-	-	-	-	-	
Fixed Angle⋆	-	-	-	-	-	-	-	
Fourier (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4	10 [40]	20 [39–144]	30 [40–100]	10
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Interp (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4	10 [40]	20 [39–144]	30 [40–100]	10
Linear Ramp (Aer)	-	-	-	-	-	-	-	
Linear Ramp (Aer)⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	74.4	-	75.2 ± 0.3	-	1 [106]	-	2 [40
TQA (Aer)	-	-	-	-	-	-	-	
TQA (Aer)⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA⋆	-	-	-	-	-	-	-	
PP								
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle	-	-	-	-	-	-	-	
Fixed Angle⋆	-	-	-	-	-	-	-	
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	74.4	-	75.2 ± 0.3	-	1 [106]	-	2 [40
TQA	-	-	-	-	-	-	-	
TQA⋆	-	-	-	-	-	-	-	
SV								
Fourier	78.8 ± 3.1	79.3 ± 3.0	74.8 ± 4.2	82.0 ± 4.1	61 [10–20]	10 [12]	699 [10–20]	570 [
Fixed Angle	78.4 ± 2.6	-	-	80.7 ± 3.1	55 [10–20]	-	-	289 [
Fixed Angle⋆	79.1 ± 2.9	-	-	81.7 ± 4.3	55 [10–20]	-	-	273 [
Interp	78.8 ± 2.9	79.3 ± 3.0	74.6 ± 3.9	81.9 ± 4.2	83 [10–20]	10 [12]	805 [10–22]	382 [
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
TQA	77.6 ± 2.6	-	71.0 ± 5.1	79.0 ± 2.9	65 [10–20]	-	592 [10–17]	280 [
TQA⋆	78.8 ± 3.0	-	74.7 ± 3.6	78.1 ± 2.1	63 [10–20]	-	584 [10–17]	60 [1
Recursive TS	78.8 ± 2.9	79.3 ± 3.0	75.0 ± 4.8	82.0 ± 4.3	81 [10–20]	10 [12]	481 [10–22]	368 [

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAXCUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

2 Tables for depth $P = 2$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle (Aer)	-	-	20 [100]	10 [40]
Fixed Angle (Aer)⋆	-	-	20 [100]	10 [40]
Fixed Angle	10 [50]	-	90 [100]	40 [30–70]
Fixed Angle⋆	10 [50]	-	90 [100]	40 [30–70]
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Interp (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp (Aer)	-	-	10 [100]	-
Linear Ramp (Aer)⋆	-	-	10 [100]	-
Linear Ramp	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA (Aer)⋆	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
		PP		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle	10 [50]	-	81 [100]	40 [30–70]
Fixed Angle⋆	10 [50]	-	81 [100]	40 [30–70]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Linear Ramp	-	-	10 [100]	-
Linear Ramp⋆	-	-	10 [100]	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
		SV		
Fourier	61 [10–20]	10 [12]	699 [10–20]	567 [10–20]
Fixed Angle	72 [10–20]	-	294 [10–22]	372 [10–20]
Fixed Angle⋆	72 [10–20]	-	293 [10–22]	360 [10–20]
Interp	83 [10–20]	10 [12]	805 [10–22]	372 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp⋆	-	-	-	-
TQA	83 [10–20]	10 [12]	804 [10–22]	361 [10–20]
TQA⋆	83 [10–20]	10 [12]	803 [10–22]	136 [10–20]
Recursive TS	81 [10–20]	10 [12]	415 [10–22]	353 [10–20]

Table 4: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	75.2 ± 4.4
Fixed Angle (Aer)	-	-	66.7 ± 2.4	78.3 ± 2.8
Fixed Angle (Aer)*	-	-	73.4 ± 1.9	79.0 ± 2.6
Fixed Angle	77.2 ± 0.9	-	79.6 ± 2.6	76.6 ± 4.8
Fixed Angle*	80.4 ± 1.8	-	79.9 ± 2.4	77.3 ± 4.4
Fourier (Aer)	70.5 ± 2.6	75.6 ± 3.8	69.5 ± 2.0	75.4 ± 2.2
Interp	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	75.6 ± 4.0
Interp (Aer)	77.5 ± 1.8	80.4 ± 2.0	74.4 ± 3.1	71.1 ± 2.5
Linear Ramp (Aer)	-	-	68.4 ± 0.5	-
Linear Ramp (Aer)*	-	-	68.4 ± 0.5	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	69.2 ± 2.0	73.1 ± 1.6	67.8 ± 1.3	73.7 ± 1.9
TQA (Aer)*	75.8 ± 1.3	80.7 ± 2.2	71.5 ± 4.2	77.3 ± 3.4
TQA	69.3 ± 1.1	73.0 ± 1.7	70.0 ± 3.9	72.8 ± 2.5
TQA*	77.4 ± 1.3	83.1 ± 1.6	79.2 ± 3.0	77.6 ± 4.0
PP				
Fourier	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.2 ± 2.5
Fixed Angle	86.9 ± 1.0	-	79.1 ± 1.6	83.6 ± 1.4
Fixed Angle*	87.8 ± 1.0	-	79.2 ± 1.6	83.6 ± 1.4
Interp	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4
Linear Ramp	-	-	81.6 ± 1.1	-
Linear Ramp*	-	-	81.6 ± 1.1	-
Recursive TS	-	-	-	-
TQA	78.9 ± 1.3	73.0 ± 1.7	70.6 ± 3.0	76.9 ± 1.9
TQA*	87.4 ± 1.2	83.1 ± 1.6	79.6 ± 2.0	83.8 ± 1.4
SV				
Fourier	75.8 ± 9.1	77.0 ± 3.9	76.4 ± 5.8	80.6 ± 8.9
Fixed Angle	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0
Fixed Angle*	85.5 ± 2.3	-	82.6 ± 2.9	88.1 ± 3.3
Interp	85.3 ± 2.4	85.5 ± 1.8	82.5 ± 4.1	86.5 ± 4.6
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	76.0 ± 3.2	73.0 ± 5.0	72.4 ± 4.2	75.7 ± 4.4
TQA*	85.5 ± 2.4	86.6 ± 2.5	82.8 ± 3.1	87.3 ± 3.3
Recursive TS	85.2 ± 2.4	87.1 ± 1.9	81.3 ± 2.8	86.1 ± 4.3

Table 5: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	R
MPS								
Fourier	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	75.2 ± 4.4	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle (Aer)	-	-	66.7 ± 2.4	78.3 ± 2.8	-	-	20 [100]	10
Fixed Angle (Aer)*	-	-	73.4 ± 1.9	79.0 ± 2.6	-	-	20 [100]	10
Fixed Angle	77.2 ± 0.9	-	79.6 ± 2.6	76.6 ± 4.8	10 [50]	-	90 [100]	40 [3
Fixed Angle*	80.4 ± 1.8	-	79.9 ± 2.4	77.3 ± 4.4	10 [50]	-	90 [100]	40 [3
Fourier (Aer)	70.5 ± 2.6	75.6 ± 3.8	69.5 ± 2.0	75.4 ± 2.2	10 [40]	20 [39–144]	30 [40–100]	10
Interp	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	75.6 ± 4.0	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Interp (Aer)	77.5 ± 1.8	80.4 ± 2.0	74.4 ± 3.1	71.1 ± 2.5	10 [40]	20 [39–144]	30 [40–100]	10
Linear Ramp (Aer)	-	-	68.4 ± 0.5	-	-	-	10 [100]	
Linear Ramp (Aer)*	-	-	68.4 ± 0.5	-	-	-	10 [100]	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	69.2 ± 2.0	73.1 ± 1.6	67.8 ± 1.3	73.7 ± 1.9	10 [40]	10 [39]	30 [40–100]	10
TQA (Aer)*	75.8 ± 1.3	80.7 ± 2.2	71.5 ± 4.2	77.3 ± 3.4	10 [40]	10 [39]	30 [40–100]	10
TQA	69.3 ± 1.1	73.0 ± 1.7	70.0 ± 3.9	72.8 ± 2.5	20 [40–50]	10 [39]	130 [40–100]	50 [3
TQA*	77.4 ± 1.3	83.1 ± 1.6	79.2 ± 3.0	77.6 ± 4.0	20 [40–50]	10 [39]	130 [40–100]	50 [3
PP								
Fourier	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.2 ± 2.5	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle	86.9 ± 1.0	-	79.1 ± 1.6	83.6 ± 1.4	10 [50]	-	81 [100]	40 [3
Fixed Angle*	87.8 ± 1.0	-	79.2 ± 1.6	83.6 ± 1.4	10 [50]	-	81 [100]	40 [3
Interp	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Linear Ramp	-	-	81.6 ± 1.1	-	-	-	10 [100]	
Linear Ramp*	-	-	81.6 ± 1.1	-	-	-	10 [100]	
Recursive TS	-	-	-	-	-	-	-	
TQA	78.9 ± 1.3	73.0 ± 1.7	70.6 ± 3.0	76.9 ± 1.9	20 [40–50]	10 [39]	130 [40–100]	50 [3
TQA*	87.4 ± 1.2	83.1 ± 1.6	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	10 [39]	130 [40–100]	50 [3
SV								
Fourier	75.8 ± 9.1	77.0 ± 3.9	76.4 ± 5.8	80.6 ± 8.9	61 [10–20]	10 [12]	699 [10–20]	567 [
Fixed Angle	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0	72 [10–20]	-	294 [10–22]	372 [
Fixed Angle*	85.5 ± 2.3	-	82.6 ± 2.9	88.1 ± 3.3	72 [10–20]	-	293 [10–22]	360 [
Interp	85.3 ± 2.4	85.5 ± 1.8	82.5 ± 4.1	86.5 ± 4.6	83 [10–20]	10 [12]	805 [10–22]	372 [
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
TQA	76.0 ± 3.2	73.0 ± 5.0	72.4 ± 4.2	75.7 ± 4.4	83 [10–20]	10 [12]	804 [10–22]	361 [
TQA*	85.5 ± 2.4	86.6 ± 2.5	82.8 ± 3.1	87.3 ± 3.3	83 [10–20]	10 [12]	803 [10–22]	136 [
Recursive TS	85.2 ± 2.4	87.1 ± 1.9	81.3 ± 2.8	86.1 ± 4.3	81 [10–20]	10 [12]	415 [10–22]	353 [

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAXCUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

3 Tables for depth $P = 3$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	30 [30–70]
Fixed Angle*	-	-	-	30 [30–70]
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Interp (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	10 [40]	30 [30–70]
TQA*	-	-	10 [40]	30 [30–70]
PP				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Fixed Angle	-	-	-	30 [30–70]
Fixed Angle*	-	-	-	30 [30–70]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	10 [40]	30 [30–70]
TQA*	-	-	10 [40]	30 [30–70]
SV				
Fourier	59 [10–19]	10 [12]	699 [10–20]	567 [10–20]
Fixed Angle	72 [10–20]	-	266 [10–20]	372 [10–20]
Fixed Angle*	72 [10–20]	-	266 [10–20]	364 [10–20]
Interp	82 [10–20]	10 [12]	803 [10–22]	372 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	803 [10–22]	357 [10–20]
TQA*	82 [10–20]	10 [12]	803 [10–22]	135 [10–20]
Recursive TS	81 [10–20]	10 [12]	415 [10–22]	357 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	78.0 ± 5.3
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	81.6 ± 6.2
Fixed Angle*	-	-	-	83.3 ± 5.2
Fourier (Aer)	80.0 ± 4.1	77.9 ± 2.7	71.6 ± 6.0	77.1 ± 2.5
Interp	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	81.4 ± 6.7
Interp (Aer)	83.4 ± 3.0	83.4 ± 3.0	82.3 ± 4.8	73.4 ± 3.8
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	82.1 ± 1.8	76.2 ± 3.9
TQA*	-	-	88.8 ± 1.1	81.5 ± 6.3
PP				
Fourier	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.6 ± 2.3
Fixed Angle	-	-	-	87.3 ± 1.3
Fixed Angle*	-	-	-	87.3 ± 1.3
Interp	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.3 ± 1.5
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	68.0 ± 1.6	80.7 ± 1.4
TQA*	-	-	79.7 ± 2.4	87.3 ± 1.3
SV				
Fourier	83.2 ± 5.9	79.2 ± 4.5	78.9 ± 7.6	88.4 ± 4.1
Fixed Angle	87.1 ± 1.8	-	87.2 ± 2.5	89.9 ± 2.2
Fixed Angle*	89.1 ± 1.9	-	87.8 ± 2.5	91.6 ± 2.7
Interp	88.8 ± 2.0	89.8 ± 1.5	87.4 ± 3.8	89.7 ± 4.1
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	81.1 ± 4.0	79.7 ± 4.8	77.3 ± 5.2	80.1 ± 5.5
TQA*	87.6 ± 2.2	88.2 ± 2.0	85.1 ± 3.8	89.9 ± 2.6
Recursive TS	88.0 ± 2.5	90.1 ± 1.8	84.9 ± 2.4	88.5 ± 3.8

Table 8: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	R
MPS								
Fourier	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	78.0 ± 5.3	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Fixed Angle (Aer)	-	-	-	-	-	-	-	
Fixed Angle (Aer)*	-	-	-	-	-	-	-	
Fixed Angle	-	-	-	81.6 ± 6.2	-	-	-	30 [3
Fixed Angle*	-	-	-	83.3 ± 5.2	-	-	-	30 [3
Fourier (Aer)	80.0 ± 4.1	77.9 ± 2.7	71.6 ± 6.0	77.1 ± 2.5	10 [40]	20 [39–144]	30 [40–100]	10
Interp	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	81.4 ± 6.7	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Interp (Aer)	83.4 ± 3.0	83.4 ± 3.0	82.3 ± 4.8	73.4 ± 3.8	10 [40]	20 [39–144]	30 [40–100]	10
Linear Ramp (Aer)	-	-	-	-	-	-	-	
Linear Ramp (Aer)*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	-	-	-	-	-	-	-	
TQA (Aer)*	-	-	-	-	-	-	-	
TQA	-	-	82.1 ± 1.8	76.2 ± 3.9	-	-	10 [40]	30 [3
TQA*	-	-	88.8 ± 1.1	81.5 ± 6.3	-	-	10 [40]	30 [3
PP								
Fourier	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.6 ± 2.3	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Fixed Angle	-	-	-	87.3 ± 1.3	-	-	-	30 [3
Fixed Angle*	-	-	-	87.3 ± 1.3	-	-	-	30 [3
Interp	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.3 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA	-	-	68.0 ± 1.6	80.7 ± 1.4	-	-	10 [40]	30 [3
TQA*	-	-	79.7 ± 2.4	87.3 ± 1.3	-	-	10 [40]	30 [3
SV								
Fourier	83.2 ± 5.9	79.2 ± 4.5	78.9 ± 7.6	88.4 ± 4.1	59 [10–19]	10 [12]	699 [10–20]	567 [
Fixed Angle	87.1 ± 1.8	-	87.2 ± 2.5	89.9 ± 2.2	72 [10–20]	-	266 [10–20]	372 [
Fixed Angle*	89.1 ± 1.9	-	87.8 ± 2.5	91.6 ± 2.7	72 [10–20]	-	266 [10–20]	364 [
Interp	88.8 ± 2.0	89.8 ± 1.5	87.4 ± 3.8	89.7 ± 4.1	82 [10–20]	10 [12]	803 [10–22]	372 [
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
TQA	81.1 ± 4.0	79.7 ± 4.8	77.3 ± 5.2	80.1 ± 5.5	83 [10–20]	10 [12]	803 [10–22]	357 [
TQA*	87.6 ± 2.2	88.2 ± 2.0	85.1 ± 3.8	89.9 ± 2.6	82 [10–20]	10 [12]	803 [10–22]	135 [
Recursive TS	88.0 ± 2.5	90.1 ± 1.8	84.9 ± 2.4	88.5 ± 3.8	81 [10–20]	10 [12]	415 [10–22]	357 [

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAXCUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

4 Tables for depth $P = 4$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	20 [100]	10 [40]
Fixed Angle (Aer)⋆	-	-	20 [100]	10 [40]
Fixed Angle	-	-	30 [100]	10 [40]
Fixed Angle⋆	-	-	30 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA (Aer)⋆	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	20 [40]
PP				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle	-	-	20 [100]	10 [40]
Fixed Angle⋆	-	-	20 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	20 [40]
SV				
Fourier	59 [10–19]	10 [12]	699 [10–20]	567 [10–20]
Fixed Angle	52 [10–20]	-	95 [10–20]	158 [10–20]
Fixed Angle⋆	52 [10–20]	-	95 [10–20]	155 [10–20]
Interp	82 [10–20]	10 [12]	803 [10–22]	372 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp⋆	-	-	-	-
TQA	83 [10–20]	10 [12]	803 [10–22]	354 [10–20]
TQA⋆	82 [10–20]	10 [12]	803 [10–22]	136 [10–20]
Recursive TS	81 [10–20]	10 [12]	414 [10–22]	352 [10–20]

Table 10: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	88.0 ± 6.0
Fixed Angle (Aer)	-	-	74.8 ± 6.2	91.2 ± 1.7
Fixed Angle (Aer)*	-	-	81.7 ± 6.6	93.1 ± 1.0
Fixed Angle	-	-	90.4 ± 1.5	90.4 ± 1.2
Fixed Angle*	-	-	90.5 ± 1.4	92.6 ± 0.8
Fourier (Aer)	74.3 ± 4.1	80.9 ± 2.7	76.9 ± 7.6	79.7 ± 3.3
Interp	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	90.5 ± 6.0
Interp (Aer)	87.8 ± 2.2	85.7 ± 3.4	88.1 ± 4.3	80.4 ± 6.6
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	68.9 ± 1.9	81.5 ± 1.0	74.7 ± 8.2	78.5 ± 2.8
TQA (Aer)*	80.7 ± 4.4	84.8 ± 0.9	77.9 ± 9.6	82.5 ± 3.4
TQA	70.5 ± 3.4	82.0 ± 1.6	84.2 ± 2.4	80.0 ± 1.8
TQA*	79.0 ± 7.0	86.9 ± 1.1	88.5 ± 1.9	88.1 ± 3.4
PP				
Fourier	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.4 ± 1.9
Fixed Angle	-	-	84.1 ± 1.4	89.9 ± 1.6
Fixed Angle*	-	-	86.1 ± 1.7	89.9 ± 1.6
Interp	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.3 ± 1.6
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	83.7 ± 1.4	82.0 ± 1.6	75.7 ± 4.8	85.4 ± 1.8
TQA*	89.2 ± 1.9	87.3 ± 1.3	83.5 ± 1.9	88.0 ± 1.5
SV				
Fourier	78.3 ± 10.6	83.6 ± 3.3	80.9 ± 8.0	90.3 ± 4.4
Fixed Angle	89.0 ± 2.2	-	91.0 ± 2.3	92.8 ± 2.2
Fixed Angle*	91.5 ± 1.7	-	91.9 ± 2.3	93.4 ± 2.2
Interp	91.3 ± 1.7	91.8 ± 1.5	90.4 ± 3.8	91.0 ± 4.8
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	84.6 ± 3.9	82.0 ± 4.9	79.8 ± 6.0	84.0 ± 5.6
TQA*	89.4 ± 1.9	89.5 ± 1.9	87.8 ± 2.7	91.1 ± 2.5
Recursive TS	90.3 ± 2.2	91.4 ± 1.7	87.6 ± 2.3	90.1 ± 3.5

Table 11: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	
MPS								
Fourier	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	88.0 ± 6.0	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Fixed Angle (Aer)	-	-	74.8 ± 6.2	91.2 ± 1.7	-	-	20 [100]	10
Fixed Angle (Aer)⋆	-	-	81.7 ± 6.6	93.1 ± 1.0	-	-	20 [100]	10
Fixed Angle	-	-	90.4 ± 1.5	90.4 ± 1.2	-	-	30 [100]	10
Fixed Angle⋆	-	-	90.5 ± 1.4	92.6 ± 0.8	-	-	30 [100]	10
Fourier (Aer)	74.3 ± 4.1	80.9 ± 2.7	76.9 ± 7.6	79.7 ± 3.3	10 [40]	20 [39–144]	30 [40–100]	10
Interp	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	90.5 ± 6.0	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Interp (Aer)	87.8 ± 2.2	85.7 ± 3.4	88.1 ± 4.3	80.4 ± 6.6	10 [40]	20 [39–144]	30 [40–100]	10
Linear Ramp (Aer)	-	-	-	-	-	-	-	
Linear Ramp (Aer)⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	68.9 ± 1.9	81.5 ± 1.0	74.7 ± 8.2	78.5 ± 2.8	10 [40]	10 [39]	30 [40–100]	10
TQA (Aer)⋆	80.7 ± 4.4	84.8 ± 0.9	77.9 ± 9.6	82.5 ± 3.4	10 [40]	10 [39]	30 [40–100]	10
TQA	70.5 ± 3.4	82.0 ± 1.6	84.2 ± 2.4	80.0 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	20
TQA⋆	79.0 ± 7.0	86.9 ± 1.1	88.5 ± 1.9	88.1 ± 3.4	20 [40–50]	10 [39]	130 [40–100]	20
PP								
Fourier	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.4 ± 1.9	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Fixed Angle	-	-	84.1 ± 1.4	89.9 ± 1.6	-	-	20 [100]	10
Fixed Angle⋆	-	-	86.1 ± 1.7	89.9 ± 1.6	-	-	20 [100]	10
Interp	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.3 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA	83.7 ± 1.4	82.0 ± 1.6	75.7 ± 4.8	85.4 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	20
TQA⋆	89.2 ± 1.9	87.3 ± 1.3	83.5 ± 1.9	88.0 ± 1.5	20 [40–50]	10 [39]	130 [40–100]	20
SV								
Fourier	78.3 ± 10.6	83.6 ± 3.3	80.9 ± 8.0	90.3 ± 4.4	59 [10–19]	10 [12]	699 [10–20]	567
Fixed Angle	89.0 ± 2.2	-	91.0 ± 2.3	92.8 ± 2.2	52 [10–20]	-	95 [10–20]	158
Fixed Angle⋆	91.5 ± 1.7	-	91.9 ± 2.3	93.4 ± 2.2	52 [10–20]	-	95 [10–20]	155
Interp	91.3 ± 1.7	91.8 ± 1.5	90.4 ± 3.8	91.0 ± 4.8	82 [10–20]	10 [12]	803 [10–22]	372
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
TQA	84.6 ± 3.9	82.0 ± 4.9	79.8 ± 6.0	84.0 ± 5.6	83 [10–20]	10 [12]	803 [10–22]	354
TQA⋆	89.4 ± 1.9	89.5 ± 1.9	87.8 ± 2.7	91.1 ± 2.5	82 [10–20]	10 [12]	803 [10–22]	136
Recursive TS	90.3 ± 2.2	91.4 ± 1.7	87.6 ± 2.3	90.1 ± 3.5	81 [10–20]	10 [12]	414 [10–22]	352

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

5 Tables for depth $P = 5$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	10 [144]	-	-
Fixed Angle (Aer)*	-	10 [144]	-	-
Fixed Angle	-	-	-	10 [40]
Fixed Angle*	-	-	-	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp (Aer)	-	10 [144]	10 [100]	-
Linear Ramp (Aer)*	-	10 [144]	10 [100]	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA (Aer)	-	10 [144]	-	-
TQA (Aer)*	-	10 [144]	-	-
TQA	-	20 [105–144]	10 [40]	10 [40]
TQA*	-	20 [105–144]	10 [40]	10 [40]
		PP		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle	-	10 [144]	-	10 [40]
Fixed Angle*	-	10 [144]	-	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp	-	10 [144]	10 [100]	-
Linear Ramp*	-	10 [144]	10 [100]	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	20 [105–144]	10 [40]	10 [40]
TQA*	-	20 [105–144]	10 [40]	10 [40]
		SV		
Fourier	58 [10–19]	10 [12]	699 [10–20]	266 [10–20]
Fixed Angle	39 [10–20]	-	50 [10–19]	121 [10–20]
Fixed Angle*	39 [10–20]	-	49 [10–19]	115 [10–20]
Interp	82 [10–20]	10 [12]	693 [10–20]	372 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	359 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]
Recursive TS	81 [10–20]	10 [12]	276 [10–20]	348 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
		MPS		
Fourier	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	90.0 ± 8.3
Fixed Angle (Aer)	-	83.3 ± 3.4	-	-
Fixed Angle (Aer)*	-	84.3 ± 3.3	-	-
Fixed Angle	-	-	-	93.8 ± 0.6
Fixed Angle*	-	-	-	95.0 ± 0.5
Fourier (Aer)	82.1 ± 6.3	82.4 ± 3.3	80.0 ± 9.5	80.5 ± 4.8
Interp	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	93.4 ± 3.6
Interp (Aer)	91.6 ± 1.6	87.2 ± 3.8	91.0 ± 3.6	86.3 ± 4.7
Linear Ramp (Aer)	-	85.7 ± 1.0	64.5 ± 2.1	-
Linear Ramp (Aer)*	-	85.7 ± 1.0	64.5 ± 2.1	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	71.1	-	83.7 ± 7.3
TQA (Aer)	-	79.2 ± 1.2	-	-
TQA (Aer)*	-	81.5 ± 1.3	-	-
TQA	-	73.4 ± 4.6	88.2 ± 1.2	82.6 ± 1.5
TQA*	-	75.2 ± 4.8	91.4 ± 1.0	90.2 ± 1.4
		PP		
Fourier	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	89.0 ± 1.5
Fixed Angle	-	88.7 ± 1.8	-	91.5 ± 1.5
Fixed Angle*	-	90.5 ± 0.8	-	91.7 ± 1.5
Interp	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.5 ± 1.8
Linear Ramp	-	88.8 ± 0.8	86.9 ± 1.3	-
Linear Ramp*	-	88.8 ± 0.8	86.9 ± 1.3	-
Recursive TS	-	88.2	-	89.0 ± 2.3
TQA	-	83.8 ± 1.1	68.6 ± 1.7	85.9 ± 1.4
TQA*	-	88.2 ± 0.8	83.5 ± 2.0	89.7 ± 1.5
		SV		
Fourier	80.0 ± 10.3	79.2 ± 6.8	81.4 ± 8.3	85.2 ± 10.7
Fixed Angle	90.0 ± 2.3	-	94.0 ± 1.8	94.9 ± 2.0
Fixed Angle*	93.2 ± 1.5	-	95.2 ± 1.7	95.3 ± 2.0
Interp	93.0 ± 1.5	92.5 ± 0.9	92.6 ± 4.1	92.5 ± 4.4
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	86.7 ± 3.7	83.7 ± 4.9	81.9 ± 6.4	86.6 ± 5.3
TQA*	91.6 ± 1.5	90.7 ± 2.0	90.6 ± 3.4	93.1 ± 2.2
Recursive TS	91.9 ± 2.1	92.6 ± 1.3	89.8 ± 3.5	91.1 ± 3.5

Table 14: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fourier	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	90.0 ± 8.3	20 [40–50]	31 [39–144]	130 [40–100]
Fixed Angle (Aer)	-	83.3 ± 3.4	-	-	-	10 [144]	-
Fixed Angle (Aer)⋆	-	84.3 ± 3.3	-	-	-	10 [144]	-
Fixed Angle	-	-	-	93.8 ± 0.6	-	-	-
Fixed Angle⋆	-	-	-	95.0 ± 0.5	-	-	-
Fourier (Aer)	82.1 ± 6.3	82.4 ± 3.3	80.0 ± 9.5	80.5 ± 4.8	10 [40]	20 [39–144]	30 [40–100]
Interp	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	93.4 ± 3.6	20 [40–50]	31 [39–144]	130 [40–100]
Interp (Aer)	91.6 ± 1.6	87.2 ± 3.8	91.0 ± 3.6	86.3 ± 4.7	10 [40]	20 [39–144]	30 [40–100]
Linear Ramp (Aer)	-	85.7 ± 1.0	64.5 ± 2.1	-	-	10 [144]	10 [100]
Linear Ramp (Aer)⋆	-	85.7 ± 1.0	64.5 ± 2.1	-	-	10 [144]	10 [100]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp⋆	-	-	-	-	-	-	-
Recursive TS	-	71.1	-	83.7 ± 7.3	-	1 [106]	-
TQA (Aer)	-	79.2 ± 1.2	-	-	-	10 [144]	-
TQA (Aer)⋆	-	81.5 ± 1.3	-	-	-	10 [144]	-
TQA	-	73.4 ± 4.6	88.2 ± 1.2	82.6 ± 1.5	-	20 [105–144]	10 [40]
TQA⋆	-	75.2 ± 4.8	91.4 ± 1.0	90.2 ± 1.4	-	20 [105–144]	10 [40]
PP							
Fourier	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	89.0 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]
Fixed Angle	-	88.7 ± 1.8	-	91.5 ± 1.5	-	10 [144]	-
Fixed Angle⋆	-	90.5 ± 0.8	-	91.7 ± 1.5	-	10 [144]	-
Interp	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.5 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]
Linear Ramp	-	88.8 ± 0.8	86.9 ± 1.3	-	-	10 [144]	10 [100]
Linear Ramp⋆	-	88.8 ± 0.8	86.9 ± 1.3	-	-	10 [144]	10 [100]
Recursive TS	-	88.2	-	89.0 ± 2.3	-	1 [106]	-
TQA	-	83.8 ± 1.1	68.6 ± 1.7	85.9 ± 1.4	-	20 [105–144]	10 [40]
TQA⋆	-	88.2 ± 0.8	83.5 ± 2.0	89.7 ± 1.5	-	20 [105–144]	10 [40]
SV							
Fourier	80.0 ± 10.3	79.2 ± 6.8	81.4 ± 8.3	85.2 ± 10.7	58 [10–19]	10 [12]	699 [10–20]
Fixed Angle	90.0 ± 2.3	-	94.0 ± 1.8	94.9 ± 2.0	39 [10–20]	-	50 [10–19]
Fixed Angle⋆	93.2 ± 1.5	-	95.2 ± 1.7	95.3 ± 2.0	39 [10–20]	-	49 [10–19]
Interp	93.0 ± 1.5	92.5 ± 0.9	92.6 ± 4.1	92.5 ± 4.4	82 [10–20]	10 [12]	693 [10–20]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp⋆	-	-	-	-	-	-	-
TQA	86.7 ± 3.7	83.7 ± 4.9	81.9 ± 6.4	86.6 ± 5.3	83 [10–20]	10 [12]	659 [10–20]
TQA⋆	91.6 ± 1.5	90.7 ± 2.0	90.6 ± 3.4	93.1 ± 2.2	81 [10–20]	10 [12]	653 [10–20]
Recursive TS	91.9 ± 2.1	92.6 ± 1.3	89.8 ± 3.5	91.1 ± 3.5	81 [10–20]	10 [12]	276 [10–20]

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

6 Tables for depth $P = 6$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	20 [100]	10 [40]
Fixed Angle (Aer)⋆	-	-	20 [100]	10 [40]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle⋆	-	-	10 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp (Aer)	-	-	10 [100]	-
Linear Ramp (Aer)⋆	-	-	10 [100]	-
Linear Ramp	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA (Aer)⋆	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	20 [40]
		PP		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle⋆	-	-	10 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp	-	-	10 [100]	-
Linear Ramp⋆	-	-	10 [100]	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	20 [40]
		SV		
Fourier	56 [10–19]	10 [12]	693 [10–20]	266 [10–20]
Fixed Angle	16 [11–17]	-	19 [10–18]	59 [10–20]
Fixed Angle⋆	16 [11–17]	-	19 [10–18]	59 [10–20]
Interp	82 [10–20]	10 [12]	686 [10–20]	372 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp⋆	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	360 [10–20]
TQA⋆	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]
Recursive TS	80 [10–20]	10 [12]	269 [10–20]	348 [10–20]

Table 16: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	92.2 ± 6.7
Fixed Angle (Aer)	-	-	85.7 ± 5.1	96.6 ± 0.5
Fixed Angle (Aer)*	-	-	89.9 ± 2.5	97.2 ± 0.2
Fixed Angle	-	-	95.1 ± 1.0	96.1 ± 0.5
Fixed Angle*	-	-	95.1 ± 1.1	96.9 ± 0.6
Fourier (Aer)	79.3 ± 7.1	83.1 ± 3.0	80.7 ± 8.8	86.1 ± 3.2
Interp	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	94.3 ± 3.6
Interp (Aer)	92.8 ± 1.3	88.3 ± 4.4	93.4 ± 2.5	89.3 ± 3.4
Linear Ramp (Aer)	-	-	65.4 ± 3.0	-
Linear Ramp (Aer)*	-	-	65.4 ± 3.0	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	74.2 ± 2.8	84.8 ± 0.9	78.6 ± 8.4	86.3 ± 2.3
TQA (Aer)*	83.3 ± 3.8	86.8 ± 1.3	84.6 ± 8.7	93.2 ± 1.1
TQA	71.7 ± 6.0	85.6 ± 1.4	88.0 ± 2.1	89.9 ± 2.7
TQA*	85.5 ± 6.7	88.8 ± 0.9	91.1 ± 1.4	93.9 ± 1.3
PP				
Fourier	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.7 ± 1.8
Fixed Angle	-	-	85.5 ± 1.6	92.2 ± 1.6
Fixed Angle*	-	-	89.1 ± 1.8	92.5 ± 1.5
Interp	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.6 ± 1.9
Linear Ramp	-	-	87.9 ± 1.3	-
Linear Ramp*	-	-	87.9 ± 1.3	-
Recursive TS	-	-	-	-
TQA	84.6 ± 1.9	85.6 ± 1.4	76.5 ± 5.9	88.0 ± 1.8
TQA*	91.3 ± 2.4	89.8 ± 1.0	85.2 ± 2.2	90.7 ± 1.6
SV				
Fourier	79.8 ± 11.1	81.3 ± 11.1	81.1 ± 9.4	84.7 ± 11.8
Fixed Angle	90.5 ± 2.5	-	95.7 ± 1.3	95.9 ± 1.3
Fixed Angle*	94.5 ± 1.5	-	96.6 ± 1.4	96.2 ± 1.4
Interp	94.2 ± 1.3	93.6 ± 1.4	93.8 ± 4.0	93.1 ± 4.9
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	88.1 ± 3.5	85.6 ± 4.4	82.4 ± 7.3	88.4 ± 5.0
TQA*	93.0 ± 1.4	91.2 ± 2.2	91.9 ± 3.1	94.2 ± 2.2
Recursive TS	93.1 ± 1.9	93.1 ± 1.3	91.0 ± 2.7	92.2 ± 3.4

Table 17: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	
MPS								
Fourier	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	92.2 ± 6.7	20 [40–50]	31 [39–144]	130 [40–100]	2
Fixed Angle (Aer)	-	-	85.7 ± 5.1	96.6 ± 0.5	-	-	20 [100]	
Fixed Angle (Aer)*	-	-	89.9 ± 2.5	97.2 ± 0.2	-	-	20 [100]	
Fixed Angle	-	-	95.1 ± 1.0	96.1 ± 0.5	-	-	10 [100]	
Fixed Angle*	-	-	95.1 ± 1.1	96.9 ± 0.6	-	-	10 [100]	
Fourier (Aer)	79.3 ± 7.1	83.1 ± 3.0	80.7 ± 8.8	86.1 ± 3.2	10 [40]	20 [39–144]	30 [40–100]	
Interp	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	94.3 ± 3.6	20 [40–50]	31 [39–144]	130 [40–100]	2
Interp (Aer)	92.8 ± 1.3	88.3 ± 4.4	93.4 ± 2.5	89.3 ± 3.4	10 [40]	20 [39–144]	30 [40–100]	
Linear Ramp (Aer)	-	-	65.4 ± 3.0	-	-	-	10 [100]	
Linear Ramp (Aer)*	-	-	65.4 ± 3.0	-	-	-	10 [100]	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	74.2 ± 2.8	84.8 ± 0.9	78.6 ± 8.4	86.3 ± 2.3	10 [40]	10 [39]	30 [40–100]	
TQA (Aer)*	83.3 ± 3.8	86.8 ± 1.3	84.6 ± 8.7	93.2 ± 1.1	10 [40]	10 [39]	30 [40–100]	
TQA	71.7 ± 6.0	85.6 ± 1.4	88.0 ± 2.1	89.9 ± 2.7	20 [40–50]	10 [39]	130 [40–100]	
TQA*	85.5 ± 6.7	88.8 ± 0.9	91.1 ± 1.4	93.9 ± 1.3	20 [40–50]	10 [39]	130 [40–100]	
PP								
Fourier	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.7 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	2
Fixed Angle	-	-	85.5 ± 1.6	92.2 ± 1.6	-	-	10 [100]	
Fixed Angle*	-	-	89.1 ± 1.8	92.5 ± 1.5	-	-	10 [100]	
Interp	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.6 ± 1.9	20 [40–50]	31 [39–144]	130 [40–100]	2
Linear Ramp	-	-	87.9 ± 1.3	-	-	-	10 [100]	
Linear Ramp*	-	-	87.9 ± 1.3	-	-	-	10 [100]	
Recursive TS	-	-	-	-	-	-	-	
TQA	84.6 ± 1.9	85.6 ± 1.4	76.5 ± 5.9	88.0 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	
TQA*	91.3 ± 2.4	89.8 ± 1.0	85.2 ± 2.2	90.7 ± 1.6	20 [40–50]	10 [39]	130 [40–100]	
SV								
Fourier	79.8 ± 11.1	81.3 ± 11.1	81.1 ± 9.4	84.7 ± 11.8	56 [10–19]	10 [12]	693 [10–20]	2
Fixed Angle	90.5 ± 2.5	-	95.7 ± 1.3	95.9 ± 1.3	16 [11–17]	-	19 [10–18]	5
Fixed Angle*	94.5 ± 1.5	-	96.6 ± 1.4	96.2 ± 1.4	16 [11–17]	-	19 [10–18]	5
Interp	94.2 ± 1.3	93.6 ± 1.4	93.8 ± 4.0	93.1 ± 4.9	82 [10–20]	10 [12]	686 [10–20]	3
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
TQA	88.1 ± 3.5	85.6 ± 4.4	82.4 ± 7.3	88.4 ± 5.0	83 [10–20]	10 [12]	659 [10–20]	3
TQA*	93.0 ± 1.4	91.2 ± 2.2	91.9 ± 3.1	94.2 ± 2.2	81 [10–20]	10 [12]	653 [10–20]	1
Recursive TS	93.1 ± 1.9	93.1 ± 1.3	91.0 ± 2.7	92.2 ± 3.4	80 [10–20]	10 [12]	269 [10–20]	3

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

7 Tables for depth $P = 7$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fourier	54 [10–18]	10 [12]	688 [10–20]	266 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	60 [10–20]
Fixed Angle*	16 [11–17]	-	18 [10–18]	59 [10–20]
Interp	82 [10–20]	10 [12]	681 [10–20]	372 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	360 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	135 [10–20]
Recursive TS	80 [10–20]	10 [12]	270 [10–20]	349 [10–20]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
		MPS		
Fourier	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	91.0 ± 7.6
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	81.3 ± 6.9	84.4 ± 3.2	81.9 ± 10.0	88.2 ± 6.1
Interp	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	95.1 ± 3.9
Interp (Aer)	94.0 ± 1.5	89.2 ± 4.7	95.0 ± 2.3	90.4 ± 2.4
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		PP		
Fourier	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	90.1 ± 1.8
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.5 ± 2.0
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		SV		
Fourier	80.0 ± 10.9	81.7 ± 10.2	81.0 ± 9.8	84.7 ± 12.2
Fixed Angle	91.0 ± 3.0	-	96.4 ± 1.0	96.8 ± 1.2
Fixed Angle*	95.5 ± 1.6	-	97.5 ± 1.1	97.2 ± 1.2
Interp	95.1 ± 1.3	94.3 ± 1.2	94.7 ± 4.2	93.3 ± 5.4
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	89.2 ± 3.4	87.0 ± 4.1	83.0 ± 7.8	89.3 ± 4.8
TQA*	93.9 ± 1.4	91.8 ± 2.2	92.5 ± 3.2	95.1 ± 1.8
Recursive TS	93.9 ± 1.9	93.8 ± 1.0	92.4 ± 4.1	92.9 ± 3.4

Table 20: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fourier	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	91.0 ± 7.6	20 [40–50]	31 [39–144]	120 [40–100]
Fixed Angle (Aer)	-	-	-	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-	-	-	-
Fixed Angle	-	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-	-
Fourier (Aer)	81.3 ± 6.9	84.4 ± 3.2	81.9 ± 10.0	88.2 ± 6.1	10 [40]	20 [39–144]	20 [40–100]
Interp	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	95.1 ± 3.9	20 [40–50]	31 [39–144]	120 [40–100]
Interp (Aer)	94.0 ± 1.5	89.2 ± 4.7	95.0 ± 2.3	90.4 ± 2.4	10 [40]	20 [39–144]	20 [40–100]
Linear Ramp (Aer)	-	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
PP							
Fourier	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	90.1 ± 1.8	20 [40–50]	31 [39–144]	120 [40–100]
Fixed Angle	-	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-	-
Interp	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.5 ± 2.0	20 [40–50]	31 [39–144]	120 [40–100]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
SV							
Fourier	80.0 ± 10.9	81.7 ± 10.2	81.0 ± 9.8	84.7 ± 12.2	54 [10–18]	10 [12]	688 [10–20]
Fixed Angle	91.0 ± 3.0	-	96.4 ± 1.0	96.8 ± 1.2	16 [11–17]	-	18 [10–18]
Fixed Angle*	95.5 ± 1.6	-	97.5 ± 1.1	97.2 ± 1.2	16 [11–17]	-	18 [10–18]
Interp	95.1 ± 1.3	94.3 ± 1.2	94.7 ± 4.2	93.3 ± 5.4	82 [10–20]	10 [12]	681 [10–20]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
TQA	89.2 ± 3.4	87.0 ± 4.1	83.0 ± 7.8	89.3 ± 4.8	83 [10–20]	10 [12]	659 [10–20]
TQA*	93.9 ± 1.4	91.8 ± 2.2	92.5 ± 3.2	95.1 ± 1.8	81 [10–20]	10 [12]	653 [10–20]
Recursive TS	93.9 ± 1.9	93.8 ± 1.0	92.4 ± 4.1	92.9 ± 3.4	80 [10–20]	10 [12]	270 [10–20]

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

8 Tables for depth $P = 8$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	10 [100]	10 [40]
Fixed Angle (Aer)*	-	-	10 [100]	10 [40]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle*	-	-	10 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	20 [40–100]	10 [40]
TQA (Aer)*	10 [40]	10 [39]	20 [40–100]	10 [40]
TQA	20 [40–50]	10 [39]	120 [40–100]	20 [40]
TQA*	20 [40–50]	10 [39]	120 [40–100]	20 [40]
		PP		
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle*	-	-	10 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	120 [40–100]	20 [40]
TQA*	20 [40–50]	10 [39]	120 [40–100]	20 [40]
		SV		
Fourier	53 [10–18]	10 [12]	681 [10–20]	266 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	50 [10–20]
Fixed Angle*	16 [11–17]	-	18 [10–18]	50 [10–20]
Interp	82 [10–20]	10 [12]	676 [10–20]	372 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	355 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]
Recursive TS	80 [10–20]	10 [12]	265 [10–20]	348 [10–20]

Table 22: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	93.8 ± 6.3
Fixed Angle (Aer)	-	-	89.5 ± 1.9	98.1 ± 0.3
Fixed Angle (Aer)*	-	-	93.6 ± 0.8	98.4 ± 0.5
Fixed Angle	-	-	96.6 ± 0.8	97.9 ± 0.4
Fixed Angle*	-	-	96.7 ± 0.8	98.3 ± 0.3
Fourier (Aer)	79.5 ± 7.1	85.2 ± 3.2	82.9 ± 9.6	86.5 ± 8.7
Interp	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	96.1 ± 3.2
Interp (Aer)	94.3 ± 1.5	90.0 ± 4.7	96.2 ± 1.6	90.8 ± 2.7
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	79.5 ± 3.5	87.5 ± 0.8	80.9 ± 7.2	91.1 ± 1.4
TQA (Aer)*	86.3 ± 1.7	89.3 ± 1.0	88.2 ± 5.9	96.0 ± 0.5
TQA	82.2 ± 6.9	88.3 ± 1.2	89.3 ± 2.0	93.5 ± 2.0
TQA*	91.1 ± 3.7	90.3 ± 1.0	92.2 ± 1.2	96.4 ± 1.0
PP				
Fourier	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.4 ± 2.2
Fixed Angle	-	-	86.2 ± 1.7	92.8 ± 1.6
Fixed Angle*	-	-	90.3 ± 1.9	93.4 ± 1.6
Interp	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.9 ± 2.1
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.2	77.4 ± 6.7	89.9 ± 1.6
TQA*	91.7 ± 2.3	91.1 ± 0.7	85.6 ± 2.5	92.0 ± 1.5
SV				
Fourier	79.0 ± 12.3	79.4 ± 11.1	79.7 ± 11.2	84.1 ± 12.8
Fixed Angle	91.8 ± 3.2	-	96.8 ± 1.1	97.2 ± 1.1
Fixed Angle*	96.4 ± 1.4	-	98.1 ± 0.9	97.8 ± 1.1
Interp	95.8 ± 1.2	94.7 ± 1.1	95.3 ± 4.0	93.7 ± 5.4
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	90.2 ± 3.5	88.1 ± 4.0	83.5 ± 8.5	90.0 ± 4.9
TQA*	94.4 ± 1.3	92.3 ± 1.8	93.2 ± 3.3	95.7 ± 1.6
Recursive TS	94.6 ± 1.8	94.3 ± 1.0	93.3 ± 4.6	93.4 ± 3.3

Table 23: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fourier	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	93.8 ± 6.3	20 [40–50]	31 [39–144]	120 [40–100]
Fixed Angle (Aer)	-	-	89.5 ± 1.9	98.1 ± 0.3	-	-	10 [100]
Fixed Angle (Aer)*	-	-	93.6 ± 0.8	98.4 ± 0.5	-	-	10 [100]
Fixed Angle	-	-	96.6 ± 0.8	97.9 ± 0.4	-	-	10 [100]
Fixed Angle*	-	-	96.7 ± 0.8	98.3 ± 0.3	-	-	10 [100]
Fourier (Aer)	79.5 ± 7.1	85.2 ± 3.2	82.9 ± 9.6	86.5 ± 8.7	10 [40]	20 [39–144]	20 [40–100]
Interp	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	96.1 ± 3.2	20 [40–50]	31 [39–144]	120 [40–100]
Interp (Aer)	94.3 ± 1.5	90.0 ± 4.7	96.2 ± 1.6	90.8 ± 2.7	10 [40]	20 [39–144]	20 [40–100]
Linear Ramp (Aer)	-	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA (Aer)	79.5 ± 3.5	87.5 ± 0.8	80.9 ± 7.2	91.1 ± 1.4	10 [40]	10 [39]	20 [40–100]
TQA (Aer)*	86.3 ± 1.7	89.3 ± 1.0	88.2 ± 5.9	96.0 ± 0.5	10 [40]	10 [39]	20 [40–100]
TQA	82.2 ± 6.9	88.3 ± 1.2	89.3 ± 2.0	93.5 ± 2.0	20 [40–50]	10 [39]	120 [40–100]
TQA*	91.1 ± 3.7	90.3 ± 1.0	92.2 ± 1.2	96.4 ± 1.0	20 [40–50]	10 [39]	120 [40–100]
PP							
Fourier	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.4 ± 2.2	20 [40–50]	31 [39–144]	120 [40–100]
Fixed Angle	-	-	86.2 ± 1.7	92.8 ± 1.6	-	-	10 [100]
Fixed Angle*	-	-	90.3 ± 1.9	93.4 ± 1.6	-	-	10 [100]
Interp	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.9 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.2	77.4 ± 6.7	89.9 ± 1.6	20 [40–50]	10 [39]	120 [40–100]
TQA*	91.7 ± 2.3	91.1 ± 0.7	85.6 ± 2.5	92.0 ± 1.5	20 [40–50]	10 [39]	120 [40–100]
SV							
Fourier	79.0 ± 12.3	79.4 ± 11.1	79.7 ± 11.2	84.1 ± 12.8	53 [10–18]	10 [12]	681 [10–20]
Fixed Angle	91.8 ± 3.2	-	96.8 ± 1.1	97.2 ± 1.1	16 [11–17]	-	18 [10–18]
Fixed Angle*	96.4 ± 1.4	-	98.1 ± 0.9	97.8 ± 1.1	16 [11–17]	-	18 [10–18]
Interp	95.8 ± 1.2	94.7 ± 1.1	95.3 ± 4.0	93.7 ± 5.4	82 [10–20]	10 [12]	676 [10–20]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
TQA	90.2 ± 3.5	88.1 ± 4.0	83.5 ± 8.5	90.0 ± 4.9	83 [10–20]	10 [12]	659 [10–20]
TQA*	94.4 ± 1.3	92.3 ± 1.8	93.2 ± 3.3	95.7 ± 1.6	81 [10–20]	10 [12]	653 [10–20]
Recursive TS	94.6 ± 1.8	94.3 ± 1.0	93.3 ± 4.6	93.4 ± 3.3	80 [10–20]	10 [12]	265 [10–20]

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

9 Tables for depth $P = 9$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		PP		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		SV		
Fourier	52 [10–18]	10 [12]	672 [10–20]	264 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	50 [10–20]
Fixed Angle*	16 [11–17]	-	18 [10–18]	50 [10–20]
Interp	82 [10–20]	10 [12]	673 [10–20]	371 [10–20]
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	658 [10–20]	356 [10–20]
TQA*	81 [10–20]	10 [12]	652 [10–20]	135 [10–20]
Recursive TS	80 [10–20]	10 [12]	264 [10–20]	352 [10–20]

Table 25: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	92.7 ± 7.4
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	79.0 ± 6.7	85.8 ± 4.0	73.6 ± 7.4	89.7 ± 3.2
Interp	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	93.1 ± 4.5
Interp (Aer)	94.1 ± 4.2	90.6 ± 4.8	95.9 ± 1.7	93.5 ± 1.9
Linear Ramp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fourier	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	91.0 ± 2.6
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fourier	78.2 ± 12.7	78.0 ± 10.8	79.1 ± 11.6	83.5 ± 13.7
Fixed Angle	92.1 ± 3.4	-	97.0 ± 1.3	97.7 ± 1.0
Fixed Angle*	97.0 ± 1.4	-	98.5 ± 0.8	98.3 ± 1.1
Interp	96.3 ± 1.1	94.8 ± 1.2	95.7 ± 4.3	93.6 ± 6.1
Linear Ramp	-	-	-	-
Linear Ramp*	-	-	-	-
TQA	90.9 ± 3.7	89.1 ± 3.8	83.7 ± 9.2	90.8 ± 4.9
TQA*	95.0 ± 1.3	93.0 ± 1.6	93.6 ± 3.3	96.2 ± 1.7
Recursive TS	95.0 ± 1.9	94.7 ± 0.9	93.6 ± 5.2	93.6 ± 3.2

Table 26: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fourier	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	92.7 ± 7.4	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle (Aer)	-	-	-	-	-	-	-
Fixed Angle (Aer)⋆	-	-	-	-	-	-	-
Fixed Angle	-	-	-	-	-	-	-
Fixed Angle⋆	-	-	-	-	-	-	-
Fourier (Aer)	79.0 ± 6.7	85.8 ± 4.0	73.6 ± 7.4	89.7 ± 3.2	10 [40]	20 [39–144]	10 [100]
Interp	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	93.1 ± 4.5	20 [40–50]	31 [39–144]	111 [40–100]
Interp (Aer)	94.1 ± 4.2	90.6 ± 4.8	95.9 ± 1.7	93.5 ± 1.9	10 [40]	20 [39–144]	10 [100]
Linear Ramp (Aer)	-	-	-	-	-	-	-
Linear Ramp (Aer)⋆	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp⋆	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-	-
TQA (Aer)⋆	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA⋆	-	-	-	-	-	-	-
PP							
Fourier	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	91.0 ± 2.6	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle	-	-	-	-	-	-	-
Fixed Angle⋆	-	-	-	-	-	-	-
Interp	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp⋆	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA⋆	-	-	-	-	-	-	-
SV							
Fourier	78.2 ± 12.7	78.0 ± 10.8	79.1 ± 11.6	83.5 ± 13.7	52 [10–18]	10 [12]	672 [10–20]
Fixed Angle	92.1 ± 3.4	-	97.0 ± 1.3	97.7 ± 1.0	16 [11–17]	-	18 [10–18]
Fixed Angle⋆	97.0 ± 1.4	-	98.5 ± 0.8	98.3 ± 1.1	16 [11–17]	-	18 [10–18]
Interp	96.3 ± 1.1	94.8 ± 1.2	95.7 ± 4.3	93.6 ± 6.1	82 [10–20]	10 [12]	673 [10–20]
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp⋆	-	-	-	-	-	-	-
TQA	90.9 ± 3.7	89.1 ± 3.8	83.7 ± 9.2	90.8 ± 4.9	83 [10–20]	10 [12]	658 [10–20]
TQA⋆	95.0 ± 1.3	93.0 ± 1.6	93.6 ± 3.3	96.2 ± 1.7	81 [10–20]	10 [12]	652 [10–20]
Recursive TS	95.0 ± 1.9	94.7 ± 0.9	93.6 ± 5.2	93.6 ± 3.2	80 [10–20]	10 [12]	264 [10–20]

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

10 Tables for depth $P = 10$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle (Aer)	-	10 [144]	10 [100]	10 [40]
Fixed Angle (Aer)*	-	10 [144]	10 [100]	10 [40]
Fixed Angle	-	-	10 [100]	11 [40–100]
Fixed Angle*	-	-	10 [100]	11 [40–100]
Fourier (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Linear Ramp (Aer)	10 [40]	30 [39–144]	44 [40–100]	10 [40]
Linear Ramp (Aer)*	10 [40]	30 [39–144]	44 [40–100]	10 [40]
Linear Ramp	10 [40]	10 [39]	-	-
Linear Ramp*	10 [40]	10 [39]	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
TQA (Aer)*	10 [40]	20 [39–144]	10 [100]	10 [40]
TQA	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
TQA*	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
		PP		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle	-	10 [144]	10 [100]	11 [40–100]
Fixed Angle*	-	10 [144]	10 [100]	11 [40–100]
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Linear Ramp	10 [40]	20 [39–144]	118 [40–100]	10 [40]
Linear Ramp*	10 [40]	20 [39–144]	118 [40–100]	10 [40]
Recursive TS	-	-	-	-
TQA	20 [40–50]	30 [39–144]	111 [40–100]	11 [40–100]
TQA*	20 [40–50]	30 [39–144]	111 [40–100]	11 [40–100]
		SV		
Fourier	50 [10–17]	10 [12]	665 [10–20]	252 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	46 [10–24]
Fixed Angle*	16 [11–17]	-	18 [10–18]	43 [10–22]
Interp	82 [10–20]	10 [12]	664 [10–20]	353 [10–20]
Linear Ramp	88 [10–20]	10 [12]	175 [10–20]	568 [10–20]
Linear Ramp*	88 [10–20]	10 [12]	175 [10–20]	568 [10–20]
TQA	83 [10–20]	10 [12]	657 [10–20]	349 [10–22]
TQA*	81 [10–20]	9 [12]	651 [10–20]	147 [10–20]
Recursive TS	79 [10–20]	9 [12]	264 [10–20]	333 [10–20]

Table 28: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	79.6 ± 6.1	77.4 ± 10.9	88.7 ± 5.6	92.8 ± 9.0
Fixed Angle (Aer)	-	85.3 ± 2.2	93.4 ± 1.7	98.8 ± 0.3
Fixed Angle (Aer)*	-	86.3 ± 2.4	96.0 ± 0.8	99.0 ± 0.4
Fixed Angle	-	-	97.4 ± 0.7	98.2 ± 1.4
Fixed Angle*	-	-	97.6 ± 0.7	98.6 ± 0.9
Fourier (Aer)	82.5 ± 7.8	85.1 ± 4.8	77.1 ± 7.3	91.6 ± 3.7
Interp	97.4 ± 1.5	86.6 ± 10.1	96.6 ± 0.9	93.8 ± 4.6
Interp (Aer)	94.3 ± 4.5	91.3 ± 4.6	96.9 ± 1.0	92.0 ± 1.6
Linear Ramp (Aer)	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8
Linear Ramp (Aer)*	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8
Linear Ramp	96.8 ± 0.5	93.6 ± 0.7	-	-
Linear Ramp*	96.8 ± 0.5	93.6 ± 0.7	-	-
Recursive TS	-	-	-	-
TQA (Aer)	85.9 ± 2.1	87.1 ± 2.5	83.7 ± 2.6	94.1 ± 0.9
TQA (Aer)*	90.1 ± 1.9	87.9 ± 2.9	88.3 ± 2.3	96.5 ± 0.6
TQA	87.7 ± 1.8	79.6 ± 9.2	90.0 ± 2.5	94.0 ± 3.3
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.6 ± 1.7	96.5 ± 1.7
PP				
Fourier	91.5 ± 1.9	91.1 ± 3.2	86.0 ± 2.2	91.3 ± 2.6
Fixed Angle	-	91.1 ± 1.4	86.7 ± 1.7	93.0 ± 1.6
Fixed Angle*	-	94.0 ± 0.5	90.7 ± 1.9	93.6 ± 1.6
Interp	94.4 ± 1.9	94.9 ± 0.8	89.6 ± 1.7	90.1 ± 2.4
Linear Ramp	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6
Linear Ramp*	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6
Recursive TS	-	-	-	-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	90.8 ± 1.6
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	92.3 ± 1.6
SV				
Fourier	77.5 ± 13.2	78.5 ± 11.7	77.9 ± 12.3	83.4 ± 13.9
Fixed Angle	92.4 ± 3.5	-	97.2 ± 1.3	98.1 ± 0.9
Fixed Angle*	97.4 ± 1.4	-	98.8 ± 0.6	98.7 ± 0.9
Interp	96.6 ± 1.1	94.8 ± 1.1	96.0 ± 3.9	94.0 ± 5.9
Linear Ramp	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8
Linear Ramp*	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8
TQA	91.3 ± 3.9	89.9 ± 3.6	83.8 ± 9.5	91.0 ± 5.1
TQA*	95.4 ± 1.4	93.2 ± 2.1	93.9 ± 3.2	96.7 ± 1.5
Recursive TS	95.3 ± 2.0	95.2 ± 0.8	93.4 ± 4.4	94.0 ± 3.2

Table 29: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fourier	79.6 ± 6.1	77.4 ± 10.9	88.7 ± 5.6	92.8 ± 9.0	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle (Aer)	-	85.3 ± 2.2	93.4 ± 1.7	98.8 ± 0.3	-	10 [144]	10 [100]
Fixed Angle (Aer)*	-	86.3 ± 2.4	96.0 ± 0.8	99.0 ± 0.4	-	10 [144]	10 [100]
Fixed Angle	-	-	97.4 ± 0.7	98.2 ± 1.4	-	-	10 [100]
Fixed Angle*	-	-	97.6 ± 0.7	98.6 ± 0.9	-	-	10 [100]
Fourier (Aer)	82.5 ± 7.8	85.1 ± 4.8	77.1 ± 7.3	91.6 ± 3.7	10 [40]	20 [39–144]	10 [100]
Interp	97.4 ± 1.5	86.6 ± 10.1	96.6 ± 0.9	93.8 ± 4.6	20 [40–50]	31 [39–144]	111 [40–100]
Interp (Aer)	94.3 ± 4.5	91.3 ± 4.6	96.9 ± 1.0	92.0 ± 1.6	10 [40]	20 [39–144]	10 [100]
Linear Ramp (Aer)	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8	10 [40]	30 [39–144]	44 [40–100]
Linear Ramp (Aer)*	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8	10 [40]	30 [39–144]	44 [40–100]
Linear Ramp	96.8 ± 0.5	93.6 ± 0.7	-	-	10 [40]	10 [39]	-
Linear Ramp*	96.8 ± 0.5	93.6 ± 0.7	-	-	10 [40]	10 [39]	-
Recursive TS	-	-	-	-	-	-	-
TQA (Aer)	85.9 ± 2.1	87.1 ± 2.5	83.7 ± 2.6	94.1 ± 0.9	10 [40]	20 [39–144]	10 [100]
TQA (Aer)*	90.1 ± 1.9	87.9 ± 2.9	88.3 ± 2.3	96.5 ± 0.6	10 [40]	20 [39–144]	10 [100]
TQA	87.7 ± 1.8	79.6 ± 9.2	90.0 ± 2.5	94.0 ± 3.3	20 [40–50]	31 [39–144]	111 [40–100]
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.6 ± 1.7	96.5 ± 1.7	20 [40–50]	31 [39–144]	111 [40–100]
PP							
Fourier	91.5 ± 1.9	91.1 ± 3.2	86.0 ± 2.2	91.3 ± 2.6	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle	-	91.1 ± 1.4	86.7 ± 1.7	93.0 ± 1.6	-	10 [144]	10 [100]
Fixed Angle*	-	94.0 ± 0.5	90.7 ± 1.9	93.6 ± 1.6	-	10 [144]	10 [100]
Interp	94.4 ± 1.9	94.9 ± 0.8	89.6 ± 1.7	90.1 ± 2.4	20 [40–50]	31 [39–144]	111 [40–100]
Linear Ramp	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6	10 [40]	20 [39–144]	118 [40–100]
Linear Ramp*	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6	10 [40]	20 [39–144]	118 [40–100]
Recursive TS	-	-	-	-	-	-	-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	90.8 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	92.3 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]
SV							
Fourier	77.5 ± 13.2	78.5 ± 11.7	77.9 ± 12.3	83.4 ± 13.9	50 [10–17]	10 [12]	665 [10–20]
Fixed Angle	92.4 ± 3.5	-	97.2 ± 1.3	98.1 ± 0.9	16 [11–17]	-	18 [10–18]
Fixed Angle*	97.4 ± 1.4	-	98.8 ± 0.6	98.7 ± 0.9	16 [11–17]	-	18 [10–18]
Interp	96.6 ± 1.1	94.8 ± 1.1	96.0 ± 3.9	94.0 ± 5.9	82 [10–20]	10 [12]	664 [10–20]
Linear Ramp	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8	88 [10–20]	10 [12]	175 [10–20]
Linear Ramp*	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8	88 [10–20]	10 [12]	175 [10–20]
TQA	91.3 ± 3.9	89.9 ± 3.6	83.8 ± 9.5	91.0 ± 5.1	83 [10–20]	10 [12]	657 [10–20]
TQA*	95.4 ± 1.4	93.2 ± 2.1	93.9 ± 3.2	96.7 ± 1.5	81 [10–20]	9 [12]	651 [10–20]
Recursive TS	95.3 ± 2.0	95.2 ± 0.8	93.4 ± 4.4	94.0 ± 3.2	79 [10–20]	9 [12]	264 [10–20]

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

Summary Tables of MIS Data

11 Tables for depth $P = 1$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	10 [20]

Table 31: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	1857.0 ± 11.0
Fixed Angle	1217.6 ± 2.6
Fixed Angle*	1857.0 ± 11.0
Interp	1857.0 ± 11.0
Linear Ramp*	-
TQA	1789.5 ± 10.7
TQA*	1857.0 ± 11.0
Recursive TS	1857.0 ± 11.0

Table 32: **Avg. Energy $\pm$ standard deviation for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)⋆	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)⋆	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA (Aer)⋆	-	-
TQA	-	-
TQA⋆	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Interp	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA	-	-
TQA⋆	-	-
SV		
Fourier	1857.0 ± 11.0	10 [20]
Fixed Angle	1217.6 ± 2.6	10 [20]
Fixed Angle⋆	1857.0 ± 11.0	10 [20]
Interp	1857.0 ± 11.0	10 [20]
Linear Ramp⋆	-	-
TQA	1789.5 ± 10.7	10 [20]
TQA⋆	1857.0 ± 11.0	10 [20]
Recursive TS	1857.0 ± 11.0	10 [20]

Table 33: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

12 Tables for depth $P = 2$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	-

Table 34: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3294.5 ± 10.2
Fixed Angle	1289.9 ± 28.1
Fixed Angle*	2494.3 ± 22.7
Interp	3294.0 ± 12.9
Linear Ramp*	-
TQA	2912.4 ± 6.2
TQA*	3315.8 ± 8.7
Recursive TS	-

Table 35: **Avg. Energy $\pm$ standard deviation for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	3294.5 ± 10.2	10 [20]
Fixed Angle	1289.9 ± 28.1	10 [20]
Fixed Angle*	2494.3 ± 22.7	10 [20]
Interp	3294.0 ± 12.9	10 [20]
Linear Ramp*	-	-
TQA	2912.4 ± 6.2	10 [20]
TQA*	3315.8 ± 8.7	10 [20]
Recursive TS	-	-

Table 36: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

13 Tables for depth $P = 3$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	-

Table 37: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3335.3 ± 47.3
Fixed Angle	929.2 ± 17.4
Fixed Angle*	3310.9 ± 10.3
Interp	3232.8 ± 18.9
Linear Ramp*	-
TQA	3260.1 ± 9.3
TQA*	3479.4 ± 5.4
Recursive TS	-

Table 38: **Avg. Energy $\pm$ standard deviation for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	3335.3 ± 47.3	10 [20]
Fixed Angle	929.2 ± 17.4	10 [20]
Fixed Angle*	3310.9 ± 10.3	10 [20]
Interp	3232.8 ± 18.9	10 [20]
Linear Ramp*	-	-
TQA	3260.1 ± 9.3	10 [20]
TQA*	3479.4 ± 5.4	10 [20]
Recursive TS	-	-

Table 39: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

14 Tables for depth $P = 4$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	-

Table 40: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3466.2 ± 8.9
Fixed Angle	1332.7 ± 20.0
Fixed Angle*	3325.0 ± 11.3
Interp	3444.1 ± 5.1
Linear Ramp*	-
TQA	3259.8 ± 3.1
TQA*	3479.2 ± 1.4
Recursive TS	-

Table 41: **Avg. Energy $\pm$ standard deviation for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)⋆	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)⋆	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA (Aer)⋆	-	-
TQA	-	-
TQA⋆	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Interp	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA	-	-
TQA⋆	-	-
SV		
Fourier	3466.2 ± 8.9	10 [20]
Fixed Angle	1332.7 ± 20.0	10 [20]
Fixed Angle⋆	3325.0 ± 11.3	10 [20]
Interp	3444.1 ± 5.1	10 [20]
Linear Ramp⋆	-	-
TQA	3259.8 ± 3.1	10 [20]
TQA⋆	3479.2 ± 1.4	10 [20]
Recursive TS	-	-

Table 42: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

15 Tables for depth $P = 5$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	-

Table 43: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3478.5 ± 5.5
Fixed Angle	1129.2 ± 9.2
Fixed Angle*	3184.5 ± 176.6
Interp	3462.5 ± 3.6
Linear Ramp*	-
TQA	3297.7 ± 5.7
TQA*	3479.0 ± 1.9
Recursive TS	-

Table 44: **Avg. Energy $\pm$ standard deviation for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	3478.5 ± 5.5	10 [20]
Fixed Angle	1129.2 ± 9.2	10 [20]
Fixed Angle*	3184.5 ± 176.6	10 [20]
Interp	3462.5 ± 3.6	10 [20]
Linear Ramp*	-	-
TQA	3297.7 ± 5.7	10 [20]
TQA*	3479.0 ± 1.9	10 [20]
Recursive TS	-	-

Table 45: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

16 Tables for depth $P = 6$

	ER	RR
MPS		
Fourier	-	-
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	-	10 [20]
Fixed Angle	-	10 [20]
Fixed Angle*	-	10 [20]
Interp	-	10 [20]
Linear Ramp*	-	-
TQA	5 [20]	10 [20]
TQA*	-	10 [20]
Recursive TS	-	-

Table 46: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	RR
MPS		
Fourier	-	-
Fixed Angle (Aer)⋆	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)⋆	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA (Aer)⋆	-	-
TQA	-	-
TQA⋆	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Interp	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA	-	-
TQA⋆	-	-
SV		
Fourier	-	3468.6 ± 13.6
Fixed Angle	-	2074.4 ± 28.2
Fixed Angle⋆	-	3390.5 ± 39.3
Interp	-	3460.2 ± 3.6
Linear Ramp⋆	-	-
TQA	1715.3 ± 1036.7	3304.4 ± 1.8
TQA⋆	-	3481.8 ± 3.1
Recursive TS	-	-

Table 47: **Avg. Energy $\pm$ standard deviation for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy		Num. Instances	
	ER	RR	ER	RR
MPS				
Fourier	-	-	-	-
Fixed Angle (Aer)⋆	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle⋆	-	-	-	-
Fourier (Aer)	-	-	-	-
Interp	-	-	-	-
Interp (Aer)	-	-	-	-
Linear Ramp (Aer)⋆	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)⋆	-	-	-	-
TQA	-	-	-	-
TQA⋆	-	-	-	-
PP				
Fourier	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle⋆	-	-	-	-
Interp	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA⋆	-	-	-	-
SV				
Fourier	-	3468.6 ± 13.6	-	10 [20]
Fixed Angle	-	2074.4 ± 28.2	-	10 [20]
Fixed Angle⋆	-	3390.5 ± 39.3	-	10 [20]
Interp	-	3460.2 ± 3.6	-	10 [20]
Linear Ramp⋆	-	-	-	-
TQA	1715.3 ± 1036.7	3304.4 ± 1.8	5 [20]	10 [20]
TQA⋆	-	3481.8 ± 3.1	-	10 [20]
Recursive TS	-	-	-	-

Table 48: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

17 Tables for depth $P = 7$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	9 [20]

Table 49: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3459.9 ± 11.8
Fixed Angle	1666.7 ± 10.3
Fixed Angle*	3237.2 ± 47.0
Interp	3462.4 ± 2.2
Linear Ramp*	-
TQA	3256.5 ± 3.3
TQA*	3492.1 ± 1.1
Recursive TS	3466.4 ± 13.2

Table 50: **Avg. Energy $\pm$ standard deviation for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)⋆	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)⋆	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA (Aer)⋆	-	-
TQA	-	-
TQA⋆	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Interp	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA	-	-
TQA⋆	-	-
SV		
Fourier	3459.9 ± 11.8	10 [20]
Fixed Angle	1666.7 ± 10.3	10 [20]
Fixed Angle⋆	3237.2 ± 47.0	10 [20]
Interp	3462.4 ± 2.2	10 [20]
Linear Ramp⋆	-	-
TQA	3256.5 ± 3.3	10 [20]
TQA⋆	3492.1 ± 1.1	10 [20]
Recursive TS	3466.4 ± 13.2	9 [20]

Table 51: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

18 Tables for depth $P = 8$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	-

Table 52: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3447.5 ± 18.4
Fixed Angle	2042.3 ± 32.5
Fixed Angle*	3253.9 ± 440.8
Interp	3447.1 ± 6.4
Linear Ramp*	-
TQA	3331.9 ± 3.3
TQA*	3461.8 ± 7.9
Recursive TS	-

Table 53: **Avg. Energy $\pm$ standard deviation for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	3447.5 ± 18.4	10 [20]
Fixed Angle	2042.3 ± 32.5	10 [20]
Fixed Angle*	3253.9 ± 440.8	10 [20]
Interp	3447.1 ± 6.4	10 [20]
Linear Ramp*	-	-
TQA	3331.9 ± 3.3	10 [20]
TQA*	3461.8 ± 7.9	10 [20]
Recursive TS	-	-

Table 54: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

19 Tables for depth $P = 9$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	-

Table 55: **Number of Instances (num. nodes in brackets) for** $P = 9$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3442.1 ± 19.5
Fixed Angle	1161.7 ± 22.8
Fixed Angle*	3032.3 ± 655.3
Interp	3441.1 ± 8.1
Linear Ramp*	-
TQA	3304.2 ± 4.3
TQA*	3465.1 ± 8.1
Recursive TS	-

Table 56: **Avg. Energy $\pm$ standard deviation for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	-	-
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	3442.1 ± 19.5	10 [20]
Fixed Angle	1161.7 ± 22.8	10 [20]
Fixed Angle*	3032.3 ± 655.3	10 [20]
Interp	3441.1 ± 8.1	10 [20]
Linear Ramp*	-	-
TQA	3304.2 ± 4.3	10 [20]
TQA*	3465.1 ± 8.1	10 [20]
Recursive TS	-	-

Table 57: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

20 Tables for depth $P = 10$

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	10 [20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	10 [20]
Linear Ramp*	-
TQA	10 [20]
TQA*	10 [20]
Recursive TS	1 [20]

Table 58: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	-
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	-
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	-
Fixed Angle	-
Fixed Angle*	-
Interp	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	3414.5 ± 35.6
Fixed Angle	2014.8 ± 26.6
Fixed Angle*	3382.9 ± 19.6
Interp	3418.3 ± 11.3
Linear Ramp*	-
TQA	3277.5 ± 2.6
TQA*	3463.6 ± 9.6
Recursive TS	3447.7

Table 59: **Avg. Energy $\pm$ standard deviation for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
	MPS	
Fourier	-	-
Fixed Angle (Aer)⋆	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Fourier (Aer)	-	-
Interp	-	-
Interp (Aer)	-	-
Linear Ramp (Aer)⋆	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA (Aer)⋆	-	-
TQA	-	-
TQA⋆	-	-
	PP	
Fourier	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Interp	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA	-	-
TQA⋆	-	-
	SV	
Fourier	3414.5 ± 35.6	10 [20]
Fixed Angle	2014.8 ± 26.6	10 [20]
Fixed Angle⋆	3382.9 ± 19.6	10 [20]
Interp	3418.3 ± 11.3	10 [20]
Linear Ramp⋆	-	-
TQA	3277.5 ± 2.6	10 [20]
TQA⋆	3463.6 ± 9.6	10 [20]
Recursive TS	3447.7	1 [20]

Table 60: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances.The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.